1. What protocol at layer 2 will you use to enable authentication on wired Ethernet network?
    a. 802.11ac
    b. NAC
    c. MAC
    **d. 802.1x**
2. A system administrator has problems with the “return-to-libc” attack (i.e., hackers can compute offsets between application and libraries in an offline manner). Which security mechanism should be deployed?
    **a. ASLR**
    b. DEP
    c. TPM
    d. KMCI
3. After investigating a network, an administrator has determined that the confidential data has been stolen by concealed within a larger picture. What method did the attacker used?
    a. OS-level
    b. Encryption
    c. Unused disk sectors
    **d. Steganography**
4. In order to protect against Brute Force Attack (BFA), which system should an administrator implement?
    a. Setuid
    **b. Key-length complexity**
    c. String sanitize
    d. Trusted Execution Environment
5. A user reports that is he cannot access a file and asks the administrator for permissions. Which of the following access control models is used?
    a. **Mandatory** (vrem ca doar administratorul sa aiba control pentru schimbarea permisiunilor)
    b. Role based
    c. Discretionary
    d. Rule-based
6. When using Data Execution Prevention (DEP) for protecting the executable space, is it possible for a user to write and execute a memory page consecutively (i.e., after modifying the NX bit)?
    a. No, DEP protects from consecutively writes and execute
    **b. Yes, only from a root/ administrator account**
    c. Yes, DEP is used to randomize memory pages that can then be writable and executable
    d. No, DEP protects from simultaneous writes and executes
7. Catherine is the data owner for the ISC company. She has asked the network administrator that she has the ability to assign read/ write privileges to her files. Which of the following systems should be deployed?
    **a. Discretionary**
    b. Role-based
    c. Mandatory
    d. Rule-based

8. When investigating a computer for possible electronic fraud, what should an investigator do first?
    a. Hire another expert for a second opinion
    **b. Make sure that all activities are logged**
    c. Shutdown the system
    d. Pray
9. Emilian receives an e-mail, but the email client says that the digital signature is not valid. Which of the following concept is broken?
    a. Availability
    b. Confidentiality
    c. **Integrity**
    d. Remediation
10. When looking for an Isolated Execution Environment, which of the following hardware platforms can be used to provide this feature?
    a. Trusted Platform Module (TPM)
    b. Transport Layer Security (TLS)
    c. Data Execution Prevention (DEP)
    **d. Intel Software Guard Extensions (SGX)**
11. Making data appear as if it is coming from somewhere other than its original source is known as what?
    **a. Phishing**
    b. Hacking
    c. Spoofing
    d. Cracking
12. To handle a large number of virtual private network (VPN) connections for a corporate network, which device would you deploy?
    a. Proxy server
    b. Load balancer
    c. Network firewall
    **d. VPN concentrator**
13. When designing a web based client-server application, with a single application server and a database cluster backend, input validation should be performed on which endpoint?
    **a. Within Javascript on client side**
    b. Not required when using HTTPS
    c. As a browser extension
    d. On the application server
14. Which of the following hardware security modules can be used to encrypt Veronica's Laptop?
    a. HIDS
    b. HSP
    c. USB
    **d. TPM**
15. Which of the following would most likely be considered for Data Loss Prevention?
    **a. USB mass storage device**
    b. Proxy server
    c. Application server content

d. Print server

16. Which type of network monitoring system employs an unacceptable traffic pattern database?
    a. Symmetric encryption based
    **b. Anomaly-based**
    c. Signature based
    d. Public key cryptography based
17. Florin sends out an email and the company wants to prove he did so. Which concept should they implement?
    a. Authenticity
    **b. Non-repudiation**
    c. Integrity
    d. Confidentiality
18. Mugur wants to use a type-1 hypervisor, which of the following defines this type of hypervisors?
    a. Application hypervisor
    b. OS-level hypervisor
    c. Hybrid hypervisor
    **d. Bare metal hypervisor**
19. Which of the following hardware security mechanism is used to allow a secure boot for most operating systems?
    **a. Trusted Platform Module (TPM)**
    b. Address Space Layout Randomization (ASLR)
    c. Intel Software Guard Extensions (SGX)
    d. ARM TrustZone
20. If one wants to reduce the attack surface of an operating system, which of the following is the best answer to accomplish this?
    **a. Update antivirus software**
    **b. Install updates**
    c. Disable unnecessary services
    d. Install a firewall
21. Which of the following cryptographic related browser settings allows an organization to access remote web server securely?
    a. 3DES
    **b. TLS**
    c. Trusted sites
    d. HMAC
22. Which characteristic of cryptographic keys makes them resistant to brute force attacks?
    a. Passphrases
    b. Encryption algorithm
    c. Complexity
    **d. Bit length**
23. What is it known as when an application runs in its own environment and does not interfere with other processes?
    a. Quarantine
    b. VPN
    c. Honeypot

    **d. Sandbox**
24. What type of device or service would you deploy to decrease the number of unwanted email messages your users receive?
    a. Firewall
    b. Antivirus software
    c. Protocol analyzer
    **d. Spam filter**
25. What is the protection mechanism deployed by modern mobile operating systems in order to isolate one process from another, and also from the operating system?
    a. Vetting process
    b. ASLR
    **c. Sandboxing**
    d. DEP
26. A web security administrator limits input fields on the web page and filters special characters on server-side. Which of the following attacks is trying the administrator to mitigate?
    a. Fuzzing
    **b. XSS** (Cross-site scripting)
    c. Spoofing
    d. Phishing
27. Florin wants to ensure that all external traffic reaches his front-end servers but also wants to protect access to his internal resources. What network design element is most suitable?
    a. VLAN
    b. NAT
    **c. DMZ**
    d. Virtualization
28. Which of the following is not part of the 3-level authentication methods?
    a. Something you are
    b. Something you own
    **c. Someone you know**
    d. Something you have
29. Which algorithm can an administrator use if he wants to verify the integrity of a downloaded file?
    a. RSA
    **b. SHA256**
    c. AES128
    d. DEP128
30. Heaps and stacks can be affected by which of the following attacks?
    a. Cross-site scripting
    **b. Buffer overflows**
    c. SQL infection
    d. rootkits